Entity-Relationship Diagram

The system utilizes a highly normalized relational schema designed for transactional integrity and analytical performance. Below is the logical representation of the enterprise data model.

```
[ Customers ] [ Suppliers ] [ Employees ] [ Users ] | | | | | 1 | 1 | (Staff)
| (Auth) | | | | | N | N +--------------------+ [ Sales ] <---------- [
Purchases ] | 1 | 1 | | | N | N [ SaleItems ] [ PurchaseItems ] | | | N | N +-
--------( 1 )------+ | [ Products ] <--- (Central Inventory Hub) | [ Expenses
] (Independent Ledger)
```

System Relational Model

Entities are interconnected through standardized Primary and Foreign Key relationships. The **Products** table acts as the central hub, linking both revenue (Sales) and capital expenditure (Purchases).

Users	PK
Id	INT [PK]
Username	NVARCHAR(100)
PasswordHash	NVARCHAR(256)
Role	NVARCHAR(50)

Employees	PK
Id	INT [PK]
Name	NVARCHAR(150)
Salary	DECIMAL(18,2)
Department	NVARCHAR(100)

Core Business Schema

Products	PK
Id	INT [PK]
Name	NVARCHAR(200)
CostPrice	DECIMAL(18,2)
SellPrice	DECIMAL(18,2)
Quantity	INT

Customers	PK
Id	INT [PK]
Name	NVARCHAR(150)
Phone	NVARCHAR(20)
Email	NVARCHAR(150)

Sales	PK
Id	INT [PK]
CustomerId	INT [FK]
Date	DATETIME
TotalAmount	DECIMAL(18,2)

SaleItems	PK
Id	INT [PK]
SaleId	INT [FK]
ProductId	INT [FK]
Quantity	INT

Operational Logic

Atomic Integrity: All Sales and Purchases are wrapped in T-SQL Transactions. This ensures that the Inventory (Products) count is decremented or incremented only when the parent record is successfully committed.

Cascading Deletes: Relation between Headers (Sales/Purchases) and Line Items (SaleItems/PurchaseItems) is maintained via `ON DELETE CASCADE`, ensuring no orphaned data remains in the system.